Interaction of Ions with Matter: Computer Simulations

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Monte Carlo simulations of ion transport (TRIM) were used to answer six questions on ion ranges, lattice damage, sputtering and dose. A 1 GeV proton beam travelled (2.9524 ± 0.0005) km in air with 41 m lateral straggling. Nuclear reactions, which the simulation omits, would remove all but about 2% of these protons before the end of range. 10 MeV protons stopped at 803 µm in beryllium. 2 MeV protons, as used in ion beam induced charge microscopy, probed 48.2 µm of silicon, and about half of the damage they created lay within the last 5 µm. At 200 keV, boron stopped 3.2 times shallower than protons in silicon and created 37 times more vacancies. The argon sputter yield of GaAs peaked between 10 and 30 keV at about 4 atoms per ion per element, close to the energy of maximum nuclear stopping. At 30 keV, gold gave 1.5 times and protons 1/740 of this yield. A 36.85 MeV proton beam placed the Bragg peak 0.12 mm short of the centre of a brain tumour 9 mm deep, where the dose was about three times that in the overlying tissue. Errors in the skin density and tumour composition entered for the simulation shift the required energy by about 1.3 MeV.

1. INTRODUCTION

An ion entering a solid loses its energy in a sequence of random collisions, so identical ions stop at different depths and leave different trails of damage. TRIM (Transport of Ions in Matter) reproduces this statistically. It follows each ion through an amorphous target, treating collisions with target nuclei as binary elastic scattering in a screened Coulomb potential and the energy lost to electrons as a continuous drag taken from tabulated stopping powers [1,2]. Displaced target atoms can be followed in turn. Averaging over thousands of ions gives the mean range, its spread, the depth profiles of ionisation and vacancies, and the number of atoms ejected from the surface.

These distributions determine the practical outcome in four applications treated in the manual [3]: the depth probed by ion beam induced charge (IBIC) microscopy, the dopant depth and lattice damage in ion implantation, the erosion rate in secondary ion mass spectrometry (SIMS), and the depth of the dose maximum in proton therapy. The report follows the six parts of the practical. Parts 1 and 2 are guided exercises on ranges; Parts 3 to 6 address the four applications.

2. THEORY

2.1 Stopping and Energy Loss

The energy lost per unit path length is the stopping power [1],

$$S = -\frac{\mathrm{d}E}{\mathrm{d}x} = S_\mathrm{e} + S_\mathrm{n}, \tag{1}$$

where S_e is the loss to target electrons (ionisation and excitation) and S_n the loss to target nuclei in elastic collisions. At high velocity S_e dominates and rises as the ion slows, so the energy deposited per unit length peaks near the end of the track (the Bragg peak). S_n becomes important only at low velocity, near the end of range for light ions. Nuclear collisions displace target atoms; a recoil with more than the displacement energy (15 eV in Si) leaves a vacancy.

The fraction of its energy E that an ion of mass M_1 can transfer to a target atom of mass M_2 in one collision is largest in a head-on collision. Conservation of energy and momentum gives [1]

$$\frac{T_\mathrm{max}}{E} = \frac{4M_1M_2}{(M_1+M_2)^2}. \tag{2}$$

Nuclear stopping is universal when expressed in the reduced energy [1]

$$\varepsilon = \frac{32.53 M_2 E}{Z_1 Z_2 (M_1+M_2)\left(Z_1^{0.23}+Z_2^{0.23}\right)}, \tag{3}$$

with E in keV, masses in atomic mass units and Z_1, Z_2 the atomic numbers of ion and target atom. The universal nuclear stopping function of Ref. [1] has its maximum at $\varepsilon = 0.33$.

2.2 Range and Straggling

The mean range R is the mean stopping depth measured along the beam axis, and the range straggling ΔR is the standard deviation of that depth. Multiple scattering also displaces each ion sideways. TRIM reports this as the mean distance of the stopped ions from the beam axis and as the lateral straggling, the standard deviation of the displacement along one transverse axis. The lateral straggling is used here for the manual's "radial straggling", the radius of the beam at the end of its range.

TABLE I. TRIM simulations. Damage modes: Quick, Kinchin–Pease estimate of damage with only the ion followed; Full, full damage cascades with every recoil followed. Densities are those entered in TRIM.

Part	Ion, energy	Target	Mode	Ions
1	H, 1 GeV	Air (N:O:Ar = 76:23:1 by atom), 1.25 mg/cm^3, 5 km	Quick	5000
2	H, 10 MeV	Be, 1.848 g/cm^3, 1 mm	Quick	5000
3	H, 2 MeV	Si, 2.321 g/cm^3, 100 µm	Full	10 000
4	B, H; 200 keV	Si; 2 µm (B), 5 µm (H)	Full	10 000 each
5	Ar, 1–1000 keV; H, Au, 30 keV	GaAs	Sputtering	5000 each
6	H, 36.85 MeV	Skin 1 mm, skull 4 mm, brain 3 mm, tumour 2 mm, brain 6 mm	Full	5000

3. METHODS

All ions entered at normal incidence. Table I lists the ion, energy, target and number of ions for each part. Parts 1 and 2 were run in SRIM-2008.04 and Parts 3, 4 and 6 in SRIM-2013.00. The version used for Part 5 was not recorded. Both differ from the SRIM-2003 release named in the manual [3].

Mean ranges, range straggling and vacancies per ion were taken from TRIM's summary output, and the radial and lateral statistics from its saved lateral data. Depth profiles of ionisation, vacancies, recoil energy and stopped ions were taken from TRIM's exported tables. TRIM labels each depth bin by its deep edge. This was confirmed with the 200 keV boron profile: its mean computed from the labels exceeds TRIM's mean range by exactly half a bin. Profiles are therefore plotted at bin centres and peaks are quoted as bin intervals. TRIM displays sputtering only as plots, so sputter yields were read from screen captures of those plots (Sec. 8).

The quoted uncertainties are Monte Carlo standard errors: $\sigma/\sqrt{N}$ for a mean and $\sigma/\sqrt{2N}$ for a standard deviation (Gaussian approximation), where σ is the straggling and N the number of ions. They describe only the finite number of simulated ions. The accuracy of TRIM's stopping-power tables, which limits agreement with real targets, was not assessed.

4. PARTICLE BEAM WEAPONS (1 GEV PROTONS IN AIR)

This part estimates where a 1 GeV proton beam fired through air stops and how wide it has become, which sets the hazard zone. At 1 GeV each collision removes only a small fraction of the proton's energy, so the range in air is kilometres. Small-angle scattering accumulates over that path and sets the beam radius.

4.1 Results

The protons travelled a mean range $R = (2.9524 \pm 0.0005)$ km, an areal density of 369 g/cm^2 at the simulated air density. At the end of the range the beam had spread sideways with a lateral straggling of (41.3 ± 0.4) m, and the stopped protons lay on average 54.5 m from the axis.

The longitudinal straggling was $\Delta R = (37.6 \pm 0.4)$ m. As a minimum safe distance, taking $R + 3\Delta R$ as the depth beyond which almost no protons travel gives (3.065 ± 0.001) km from the source, about 110 m beyond the mean range.

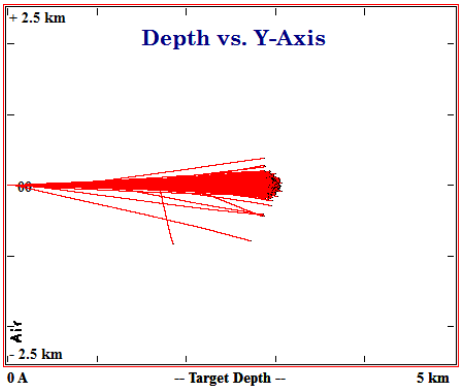


FIG. 1. Trajectories of 1 GeV protons in air projected onto the depth–transverse plane, as displayed by TRIM (depth 0–5 km, transverse ±2.5 km).

Within TRIM's model the end of the range is more hazardous than the region near the source. There the energy deposited per unit path rises to the Bragg peak (Sec. 2.1) and the beam envelope is at its widest (Fig. 1). The radial spread is set by rare large-angle collisions as well as by many small deflections: the few protons that stopped early lie up to 1.1 km off axis (Fig. 2).

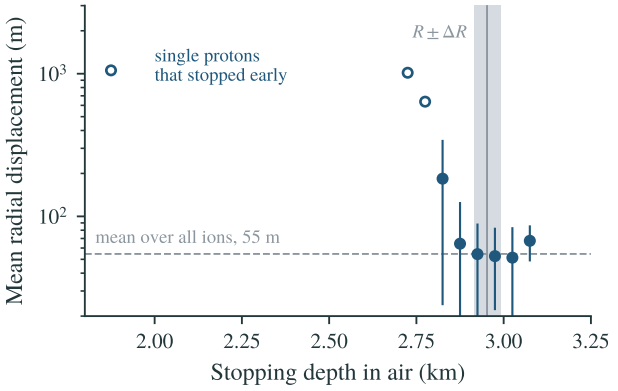


FIG. 2. Mean radial displacement of the 1 GeV protons stopped in each 50 m depth bin (5000 ions; logarithmic scale). Bars show the radial straggling within each bin; open circles are bins holding a single proton. The shaded band marks $R \pm \Delta R$ and the dashed line the mean radial displacement of all ions. Most protons stopped within 2.85–3.1 km, 50–70 m from the axis; the three that stopped earlier had been deflected 0.6–1.1 km.

4.2 Discussion

The safe distance is set by R; ΔR adds only 4% to it. The lateral straggling, 1.4% of R, gives a hazard zone near the end of range about 80 m across.

TRIM omits nuclear reactions, and at 1 GeV these dominate. The nuclear interaction length of air is 90.1 g/cm^2 [4], or 720 m at the simulated density. Over 369 g/cm^2 only $\exp(-369/90.1) = 1.7\%$ of protons would reach the end of range without a nuclear interaction. The rest would produce secondary neutrons and charged particles, mostly within the first kilometre, and neutrons can travel beyond R. The answers to (c) and (d) therefore describe the primary protons only. The simulated air density, 1.25 mg/cm^3, is about 4% above that of dry air at 20 °C; because R scales inversely with density, R is uncertain by the same fraction for real air. A hadronic transport calculation, which includes nuclear reactions and secondary particles, is needed before (c) can be used as a safety distance.

5. PROTON RANGES IN BERYLLIUM (10 MEV)

This part repeats the range calculation in a dense solid. Be has about 1500 times the density of air, so the same kinds of collisions stop a proton within a millimetre.

5.1 Results

The 10 MeV protons stopped at $R = (803.3 \pm 0.2)$ µm with a range straggling $\Delta R = (11.7 \pm 0.1)$ µm, 1.5% of R. The lateral straggling was (13.0 ± 0.1) µm, and the stopped protons lay on average 16.1 µm from the axis.

Fig. 3 shows the trajectories in the 700–800 µm window, which, as the manual notes, covers only the end of the range [3].

5.2 Discussion

10 MeV protons in Be stop within a depth interval of a few tens of micrometres at 0.8 mm, and the lateral straggling (13 µm) is comparable to ΔR. The relative straggling (1.5%) and radial spread (1.6%) are close to those of Part 1, as expected because both are set by the same kinds of collisions. The quick calculation followed only the ion, which is sufficient for ranges but gives no damage profile. Setting the plot window to the full 0–1 mm depth would show the whole track, including the narrow entrance region.

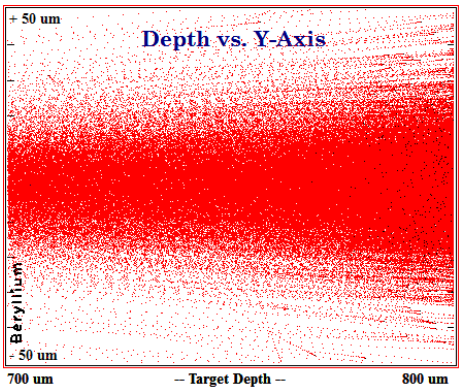


FIG. 3. Trajectories of 10 MeV protons in Be in the end-of-range window, depth 700–800 µm and transverse ±50 µm, as displayed by TRIM.

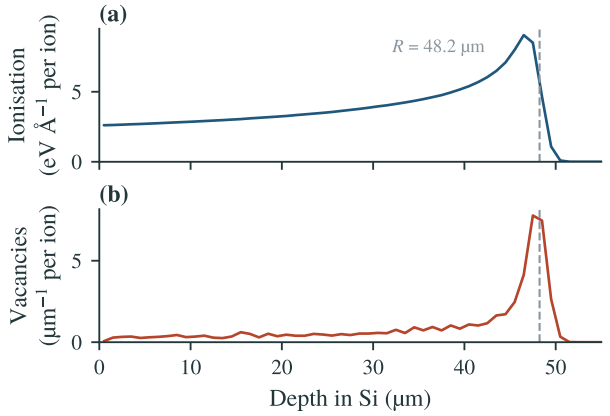


FIG. 4. Depth profiles for 2 MeV protons in Si (10 000 ions): (a) energy deposited in ionisation by the protons and (b) Si vacancies produced. The dashed line marks the mean range R. Ionisation peaks in the 46–47 µm bin and vacancies in the 47–48 µm bin.

6. NUCLEAR MICROSCOPY (IBIC IN SILICON, 2 MEV PROTONS)

In IBIC microscopy, a focused proton beam generates electron–hole pairs along its track, and the collected charge maps the electrically active regions of a buried device. The depth probed is the proton range. The same protons create defects that trap charge, so the depth where damage concentrates limits how long a region can be imaged.

6.1 Results

The depth probed by 2 MeV protons is their range, $R = (48.24 \pm 0.01)$ µm, with $\Delta R = (1.189 \pm 0.008)$ µm. The lateral straggling was (2.01 ± 0.01) µm, and the stopped protons lay on average 2.52 µm from the axis.

Vacancy production peaked in the 47–48 µm bin, just before the mean range (Fig. 4 b). The peak density, 7.8 vacancies per µm per ion, was 17 times that at 25 µm, and 48% of the 50.6 vacancies per ion lay between 45 and 50 µm. The proton's nuclear stopping rises as it slows, so near the end of range more collisions transfer more than the 15 eV displacement energy. The energy transferred to Si recoils peaked in the same bin.

Fig. 5 shows the trajectories. Ionisation, the source of the IBIC signal, rises gradually along the track and peaks 1 µm shallower than vacancy production (Fig. 4). Ionisation by recoiling Si atoms was 2.7×10^{-4} of that by the protons and is not plotted.

6.2 Discussion

IBIC signal is generated over the full 48 µm range, while 48% of the damage is created within the last 5 µm. Structures at mid-range, where the damage is about 0.5 vacancies per µm per ion at 25 µm, can therefore be imaged for longer than structures near the end of range. TRIM counts every displaced atom as a permanent vacancy in an amorphous target at 0 K. Real Si recombines a large fraction of Frenkel pairs at room temperature,

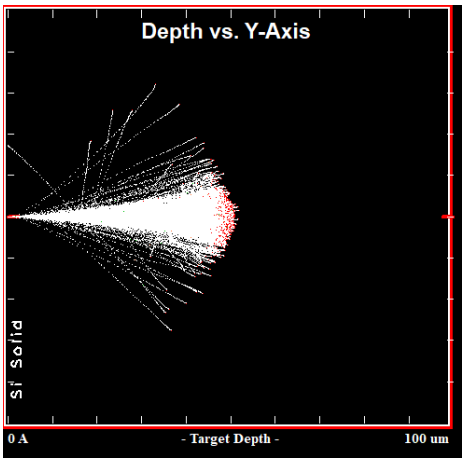


FIG. 5. Trajectories of 2 MeV protons in Si (depth 0–100 µm), as displayed by TRIM.

so the absolute vacancy count is an upper bound, while the depth of the damage peak follows from the stopping and is more reliable. Converting vacancies into a charge-collection loss requires a measured defect-trapping coefficient. Measuring IBIC signal loss against proton fluence on a test device would supply it.

7. IMPLANTATION (200 KEV B IN SI)

Implanted boron makes silicon p-type. Its depth profile sets the junction depth, and the damage it causes must be removed by annealing. This part compares 200 keV boron with 200 keV protons.

7.1 Results

Boron stopped at $R = (0.569 \pm 0.001)$ µm with $\Delta R = (0.1053 \pm 0.0007)$ µm; protons at (1.849 ± 0.001) µm with (0.1101 ± 0.0008) µm. Protons therefore penetrated 3.2 times deeper with almost the same straggling, so the boron profile is about three times broader relative to its depth (Fig. 6 a). The boron profile peaks at 0.60–0.62 µm, deeper than its mean, consistent with a tail of ions that were scattered through large angles and stopped at shallower depths.

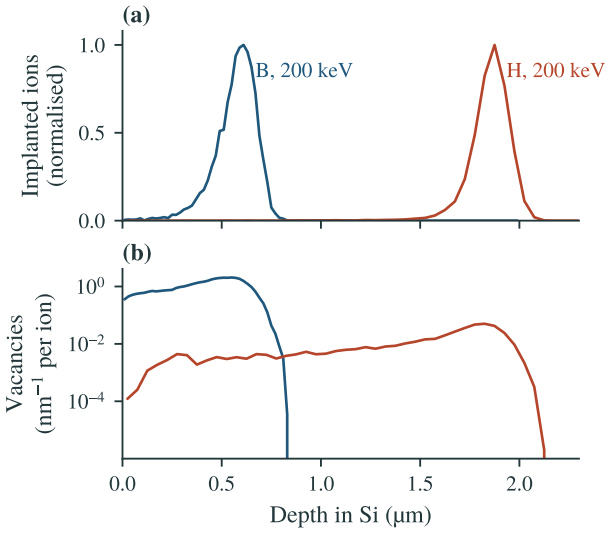


FIG. 6. 200 keV boron (blue) and hydrogen (rust) in Si, 10 000 ions each: (a) depth distribution of the implanted ions, each normalised to its maximum; (b) Si vacancies per nm per ion, on a logarithmic scale.

Boron produced 795 vacancies per ion and protons 21.3, a ratio of 37. In a single collision with Si, Eq. (2) gives a maximum energy transfer of 81% of the ion energy for boron but 13% for hydrogen, so boron produces more energetic recoils that displace further atoms in cascades. The boron damage peaks at 0.55 µm, shallower than the implanted boron at 0.61 µm; the proton damage peaks at 1.83 µm, just before the peak of the proton distribution at 1.88 µm (Fig. 6 b).

7.2 Discussion

For doping, boron places the acceptors within 0.57 µm with a 0.11 µm spread, at the cost of 37 times more damage per ion, concentrated in the doped layer itself. The vacancy counts are upper bounds, because TRIM does not allow defects to recombine. TRIM treats Si as amorphous. In crystalline Si, ions aligned with a lattice axis channel along it and form a deep tail beyond the profile in Fig. 6 a. Simulating the implant with a binary-collision code that includes the crystal lattice, at the tilt angle used in practice, would test whether the profile here is adequate for the wafer.

8. SPUTTERING (SIMS ON GAAS WITH AR)

In SIMS and focused-ion-beam machining, the sputter yield is the number of target atoms ejected per incident ion. Sputtering requires the collision cascade to deliver energy to atoms within a few atomic layers of the surface. Its yield is approximately proportional to the nuclear stopping at the surface [5]. For Ar on Ga and As, Eq. (3) places the maximum of nuclear stopping at 37 and 38 keV.

8.1 Results

The yield at each energy was read from TRIM's integral plot of atoms reaching the surface against their energy normal to the surface. Atoms with less than the 1.6 eV surface barrier marked on the plot are not sputtered, so the yield is the integral curve just above 1.6 eV. Pixel positions were converted to yield using the plot gridlines; one pixel corresponds to less than 0.03 atoms per ion for Ar. The Ga and As curves coincide to within the line width for Ar and Au. For H, the larger Ga value is quoted. The yields are therefore per element.

The Ar yield rose from 2.2 atoms per ion at 1 keV to 4.1 at 10 keV and 4.2 at 30 keV, then fell to 1.1 at 1000 keV (Fig. 7). The 30 keV value exceeds the 10 keV value by only 0.1 atoms per ion (2%). The statistical uncertainty of the yields could not be estimated from the plots, so the data place the maximum between 10 and 30 keV rather than at 30 keV.

At the same 30 keV, protons gave 0.006 atoms per ion, 1/740 of the Ar yield, and Au gave 6.1 atoms per ion, 1.5 times the Ar yield.

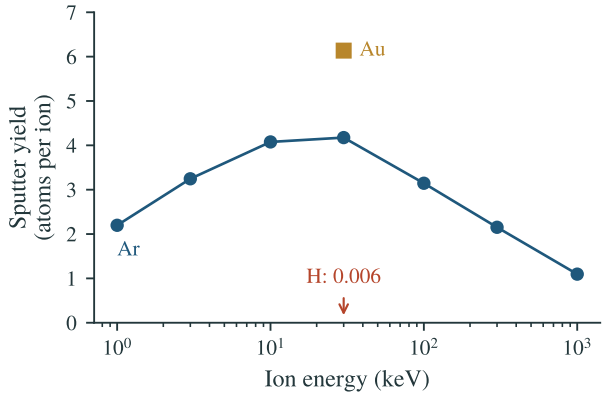


FIG. 7. Sputter yield of GaAs per element against ion energy for Ar (circles), and for Au (square) and H at 30 keV, read from TRIM's integral sputtering plots at the 1.6 eV surface barrier. The H yield is too small to show on this scale.

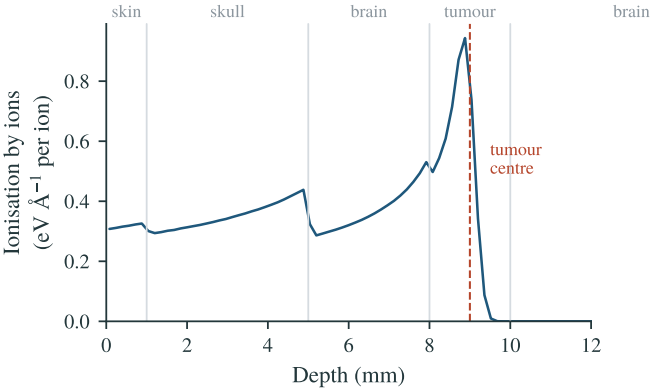


FIG. 8. Ionisation by 36.85 MeV protons against depth through the layered target (5000 ions), with the layer boundaries and the tumour centre (dashed) marked. The steps at 1, 5 and 8 mm reflect the densities and compositions entered for the layers. Recoil ionisation is negligible on this scale and is not shown.

8.2 Discussion

The Ar yield maximum lies at or just below the 37–38 keV maximum of nuclear stopping, consistent with the linear-cascade picture [5]. At lower energy less energy is available to the cascade. At higher energy electronic stopping takes a larger share, and the nuclear energy is deposited deeper, out of reach of the surface. The ordering H $\ll$ Ar $<$ Au follows the nuclear stopping, which increases with ion mass and atomic number. A proton transfers little energy per collision (Eq. (2)) and deposits most of its energy in electronic stopping. The yield is plotted per element, and the total yield of GaAs is about twice the plotted value. Exporting TRIM's sputtered-atom file, rather than reading plots, would give the total yield, the separate Ga and As yields (preferential sputtering changes the surface composition seen by SIMS) and a statistical uncertainty. With these, additional runs at 15–60 keV could locate the maximum.

9. ION BEAM RADIATION TREATMENT (PROTONS)

Proton therapy exploits the Bragg peak: a proton deposits most of its energy near the end of its range, so choosing the energy places the dose maximum in the tumour while the overlying tissue receives less. The target modelled skin (1 mm), skull (4 mm) and brain (3 mm) above a 2 mm tumour, so the tumour centre lies 9.0 mm below the skin [3]. The manual assumes negligible beam broadening.

9.1 Results

At 36.85 MeV the ionisation by the protons peaked in the 8.80–8.96 mm bin, inside the tumour but (0.12 ± 0.08) mm short of its centre (Fig. 8). The protons stopped at $R = (9.031 \pm 0.003)$ mm with $\Delta R = 0.21$ mm. The mass stopping power at entry, 16.0 MeV cm^2/g, converts one 0.16 mm bin of tissue into 0.27 MeV of beam energy. An energy higher by (0.2 ± 0.1) MeV would therefore centre the peak.

Dose is energy per unit mass, so the linear ionisation was divided by each layer's density before comparing tumour and overlying tissue. Interpolating between bin centres, the dose at the tumour centre was 3.1 times the mean dose in the skin, skull and brain traversed (0–8 mm). Moving the centre by half a bin either way changes the ratio to 2.5 and 3.5, because the centre lies on the steep distal edge of the peak. At the Bragg peak itself the ratio was 3.7. Without the density correction the ratio of linear ionisation was 2.2 at the centre.

Fig. 9 shows the trajectories through the layers. The energy transferred to recoils, the manual's measure of damage, also peaked in the 8.80–8.96 mm bin when averaged over three bins.

9.2 Discussion

The simulation demonstrates the dose advantage: at the centre of a tumour 9 mm deep, the dose was about three times the mean dose above it. The result depends on two errors in the target definition. The skin was entered at 1.92 g/cm^3, the skull value, whereas skin is soft tissue with a density close to that of water, about 1.1 g/cm^3. The extra 0.08 g/cm^2 shifts the peak about 0.8 mm shallower than in a correct target, equivalent to 1.3 MeV of beam energy. The tumour and the brain beyond it were entered with brain mass fractions in place of atomic fractions, reducing hydrogen from 63% to 10% of the atoms. This lowers the electron density per gram by 8.5%, and the ionisation drops

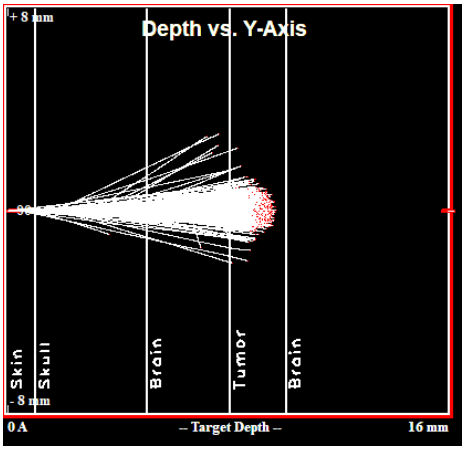


FIG. 9. Trajectories of 36.85 MeV protons through the layered target (depth 0–16 mm, transverse ±8 mm), as displayed by TRIM, with the layer boundaries marked.

by 12% at the brain–tumour boundary (Fig. 8). For the 0.9 mm travelled in the tumour, this moves the peak about 0.1 mm deeper. Re-running with the tabulated compositions and densities of skin and brain, and then adjusting the energy in steps of about 0.1 MeV until the peak bin contains 9.0 mm, would give a defensible energy. The lateral straggling, (0.247 ± 0.002) mm, is a quarter of the tumour radius, so the manual's neglect of beam broadening holds for this geometry. Covering the whole 2 mm tumour would require several energies, which spreads the peak and lowers the dose ratio.

10. CONCLUSION

Monte Carlo statistics were not limiting in any part: all standard errors are at most 1% of the quantity. The results are limited by the physics TRIM omits and by the target definitions. A 1 GeV proton travels (2.9524 ± 0.0005) km in air with (41.3 ± 0.4) m lateral straggling. Only 1.7% of protons would reach that depth without a nuclear reaction, which TRIM omits, so a hadronic transport calculation is needed for a safety distance. 10 MeV protons stop at (803.3 ± 0.2) µm in Be. 2 MeV protons probe (48.24 ± 0.01) µm of Si, and vacancy production peaks at 47–48 µm, with 48% of the damage in the last 5 µm. At 200 keV, boron stops 3.2 times shallower than protons and creates 37 times more vacancies; channelling in crystalline Si, absent from TRIM, should be simulated before the profile is applied to a wafer. The Ar sputter yield of GaAs peaks between 10 and 30 keV at about 4 atoms per ion per element, close to the nuclear-stopping maximum. Exporting sputtered-atom data would resolve the 2% difference between 10 and 30 keV. A 36.85 MeV proton beam places the Bragg peak (0.12 ± 0.08) mm short of the centre of a tumour 9 mm deep, where the dose is 3.1 (2.5–3.5) times that in the tissue above. Correcting the skin density and the tumour composition would lower the required energy by about 1.3 MeV.

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